UU-T-120G
October 31, 1980
SUPERSEDING
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October 31, 1967

FEDERAL SPECIFICATION

TAPE, TELETYPEWRITER, PERFORATOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of unpunched teletypewriter perforator paper tape in roll form.

1.2 Classification. Teletypewriter perforator tape in accordance with this specification shall be of the following two types.

Type I - Oiled tape.
Type II - Unoiled tape (for optical readers).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Description. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specification required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth, Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply:

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Technical Association of the Pulp and Paper Industry (TAPPI):

T 401 - Fiber Analysis of Paper and Paperboard.
T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Product.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 411 - Thickness (Caliper) of Paper and Paperboard.
T 413 - Ash in Paper.
T 414 - Internal Tearing Resistance of Paper.
T 425 - Opacity of Paper.
T 435 - Hydrogen Ion Concentration (pH) of Paper Extracts - Hot Extraction Method.
T 494 - Tensile Breaking Properties of Paper & Paperboard (Using Constant Rate of Elongation Apparatus).

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Stock. The teletypewriter perforator paper tape shall be made from 100 percent bleached chemical wood pulp and shall conform to the requirements in table I when tested in accordance with 4.4.

3.1.2 Impregnating oil. The oil used in fabricating type I tape shall be well refined, paraffin base, light grade lubricating oil. It shall be light in color, free from acids, and as free of odor as commercially possible. The oil shall be certified in accordance with 4.3.

3.2 Physical properties. The end item tapes and their components, as applicable, shall conform to the requirements of table I.

Table I. Physical properties

Characteristic	Type I and II	Test method
Base paper:		
Basis weight (24 x 36 - 500), pounds (± 5%)	50	TAPPI T 410
Tearing resistance, grams		
Machine direction	80 1/ 55 2/	TAPPI T 414
Cross machine direction	90 1/ 55 2/	
Opacity (type II only), percent, minimum	35	TAPPI T 425
Acidity, pH, minimum	4.5	TAPPI T 435
Ash, percent, maximum	1.0	TAPPI T 413
Grit content, percent, maximum	0.04	4.6.2
End item tape:		
Thickness, inch (± 0.0003), average	0.0040	TAPPI T 411
Tensile breaking strength (machine direction), pounds per inch, minimum	35	TAPPI T 404 3/

1/ Maximum.
2/ Minimum.
3/ TAPPI T 494 may be used in lieu of TAPPI T 404.

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3.3 Oil content. Type I tape shall be uniformly impregnated with oil conforming to 3.1.2 to form an oiled tape containing not less than 8 nor more than 16 percent oil, based on the weight of the deoiled paper when tested in accordance with 4.6.1.

3.4 Color. Type I tape shall be buff, yellow, red, white, or green, as specified (see 6.2). Type II tape shall be black. The color of type I and type II tape shall be uniform throughout the roll.

3.5 Roll assembly. Perforator teletypewriter tape shall be in rolls of one continuous length without splice. The tape at the inner end of the roll shall be fastened to the core so that it cannot be pulled loose without tearing. The outer end of the tape shall be fastened to the adjacent layer in such a way that it is retained in shipment and handling but can be pulled loose without cutting and without damaging more than two turns of the roll.

3.5.1 Roll diameter. The outside diameter of the roll shall be 8, 11-1/2, 12, or 14 inches, with a tolerance of plus or minus 1/8 inch, as specified (see 6.2).

3.5.2 Roll length. When the outside diameter of the roll is not specified, the minimum length required shall be specified (6.2), and no roll shall contain less than the specified minimum length.

3.5.3 Roll width. Rolls of perforator teletypewriter tape shall be of the following widths, in inches, 0.686, 0.875, and 1.000, with a tolerance of plus or minus 0.003 inch, as specified (see 6.2).

3.5.4 Warning signal. Each roll of tape shall be marked with a contrasting color on at least one-half the width of the tape. This mark shall begin 20 to 30 ft from the inner (core) end of the roll and shall extend to within 5 to 10 ft of the inner end of the roll.

3.5.5 Roll core. Each roll of perforator teletypewriter tape shall be uniformly and tightly wound on a fiberboard core. The core shall have a wall thickness between 1/8 and 3/16 inch. The length of the core shall be equal to the specified tape width with a tolerance of plus or minus 1/16 inch. Cores shall have an inside diameter of either 0.95 to 1.05 inch, or 1.9 to 2.1 inch, as specified (see 6.2).

3.6 Workmanship. The teletypewriter perforator tape shall be free from translucent blemishes, slime spots, wrinkles, holes, cuts, tears, splices, folds or creases. The tape rolls shall be cleanly cut, without nicks or ragged edges, and with smooth, undamaged roll faces. The rolls shall be free from paper dust, lint, and fuzz. The tape shall unwind without sticking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Sampling. Sampling for inspection and testing shall be performed in accordance with MIL-STD-105, except when otherwise indicated.

4.3 Certification. A certificate shall be submitted with each bid specifying that the impregnating oil will conform to the requirements in 3.1.2.

4.4 Testing of paper component. Each lot of paper used in the fabrication of teletypewriter perforator tape shall be tested for the characteristics and in accordance with the referenced test methods in table II. The lot size shall be expressed in pounds of base paper received from one source of supply, and each sample test unit shall consist of paper manufactured by the same source, under essentially the same conditions and identified as used in the production of the quantity of tape submitted for inspection at one time. The test sample unit shall consist of 3,000 in^2 of paper. The inspection level shall be S-1. Test results shall include all values upon which the results are based and there shall be no failure to meet the requirements.

Table II. Testing of base paper content

Characteristic	Specification reference	
	Requirement	Test method
Fiber identification	3.1.1	TAPPI T 401
Basic weight (24 x 36 - 500), pounds	Table I	TAPPI T 410
Tearing resistance, grams, machine direction and cross machine direction	Table I	TAPPI T 414
Opacity (type II), percent	Table I	TAPPI T 425
Ash, percent	Table I	TAPPI T 413
Acidity, pH	Table I	TAPPI T 435
Grit content, percent	Table I	4.6.2

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4.5 Inspection of the end item.

4.5.1 Examination. The end item shall be examined for the defects defined in 4.5.2, 4.5.3, 4.5.4, and 4.5.5, using the inspection level and acceptable quality level (AQL) specified therein. The sample unit for examination of roll defects shall be one roll, and not more than two rolls shall be selected from any one package.

4.5.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.5.3 Examination for visual defects. The rolls of teletypewriter perforator tape shall be examined for the defects listed below. The inspection level shall be S-3 with an AQL of 1.5 percent defective.

Examine	Defect
Color	Not color specified. Color not uniform throughout roll.
Tape on roll	Not uniformly and tightly wound on roll; roll telescoped. Faces of roll not smooth and undamaged. Not wound on fiber core; core crushed, broked, mutilated, or collapsed.
Appearance	Surfaces not clean; presence of dirt or foreign matter. Presence of translucent blemishes, cuts, holes, tears, creases, or wrinkles. Presence of paper dust, lint, fuzz, or slime spots. Oil, as applicable, not uniformly applied to tape; not covering entire area.
Construction	Outer end of tape glued in such a manner that more than two layers of tape are damaged at start of unwinding. Inner end of tape not securely attached to core; loose; does not tear fibers when detached. Does not unroll evenly and uniformly; tape sticks, causing tears. Tape wound unevenly, causing wrinkles or creases within the roll. Roll not continuous; any splice, joint, seam, tear, or break within the roll.
Warning signal	Warning signal not in a contrasting color to color of the tape. Width of color signal less than one-half tape width. Warning signal omitted from inner end. Does not start within 5 to 10 feet of core; does extend for a distance of 10 to 25 ft.

4.5.4 Examination for dimensional defects. The rolls of teletypewriter perforator tape shall be examined for the dimensional defects listed below. For this inspection, no more than one sample unit shall be selected from a shipping container. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

Examine	Defect
Diameter of roll	Outside diameter not within ± 1/8 inch of dimension specified.
Length of core	Not within plus or minus 1/16 inch of tape width.
Thickness of core wall	Greater than 3/16 inch or less than 1/8 inch.
Inside diameter of core	Inside diameter of core not within specified dimensions.

4.5.5 Examination of length. Te rolls of teletypewriter perforator tape shall be examined for length as applicable. The roll length shall be not less than specified. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

4.5.6 Testing. The roll of teletypewriter perforator tape shall be tested for the characteristics and in accordance with the referenced test methods in table III. The lot size shall be expressed in rolls of tape of one type, size, and color. The test sample unit shall consist of one roll of tape, and no more than one sample unit shall be taken from any one shipping container. The inspection level shall be S1. Test results shall include all values upon which the results are based, and there shall be no failure to meet the requirements. The first three layers of tape shall be removed from the rolls before taking specimens for tests.

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Table III. Testing of the end item

Characteristic	Specification reference	
	Requirement	Test method
Thickness	Table I	TAPPI T 411
Tensile breaking strength, machine direction 1/	Table I	TAPPI T 404 2/
Oil content, type I only	3.3	4.6.1
Width, roll	3.5.3	4.6.3

1/ Use the entire width of tape up to 1 inch.
2/ TAPPI T 494 may be used in lieu of TAPPI T 404.

4.6.1 Oil content. Perform two determinations on the sample unit. Cut the tape into 6 x 40 mm strips and allow to condition as specified in TAPPI T 402. Weigh out 10 ± 0.01 g of the strips. Make small zigzag folds in the strips, then place them lengthwise into the siphon cup of a Soxhlet extractor. Do not pack the strips tightly into the siphon cup. Fill the siphon cup with carbon tetrachloride to cover the sample strips. Allow the liquid to stand for 5 minutes without draining. Then, add 1.5 times the volume of carbon tetrachloride required to fill the siphon cup. Extract by heating until the siphon cup has been filled by condensation and emptied by siphon 30 times. Remove the sample strips as specified in TAPPI T 402. Weigh the reconditioned sample strips to the nearest 0.01 g. Calculate the percent oil as follows:

$$\text{Percent oil content} = \frac{\text{Original sample weight - extracted sample weight}}{\text{Extracted sample weight}} \times 100$$

Report the percent oil content as the average of the two determinations to the nearest 0.1 percent.

4.6.2 Grit content. Perform two determinations on the sample unit. Condition the sample as specified in TAPPI T 402. Weigh out about 25 g of the conditioned tape to the nearest 0.01 g and place in an 800-ml Kjeldahl flask. Add 100 ml of fuming HNO_3. Warm until the paper becomes a thin paste, then boil gently until a low viscosity solution remains. Evaporate the solution to a volume of 30 to 40 ml, then cool to room temperature. Add 20 ml of concentrated H_2SO_4 and heat until white fumes of SO_3 are evolved. Clarify the solution by adding small volumes of fuming HNO_3 allowing the drops to run down the inside wall of the flask and reheat until SO_3 fumes are evolved. Repeat this clearing operation as often as is necessary to produce a clear light straw color solution which does not darken after further heating. When clear, give the solution a final fuming to remove the HNO_3. Cool the solution to room temperature and dilute to 400 ml with distilled water. Boil gently for 10 minutes. Filter through a previously dried and weighed medium fritted glass crucible. Rinse the flask three times with distilled water, and, after each rinse, pour the water through the fritted crucible. Dry the crucible with contents at $105^{\circ} \pm 2^{\circ}$ C, then cool to room temperature and weigh. Calculate the percent of grit as follows:

$$\text{Percent of grit} = \frac{\text{Weight of solids after drying}}{\text{Original weight of sample}} \times 100$$

Caution: Because of the corrosive acid fumes, the digestion must be conducted in a well ventilated hood. Extreme caution should be used when working with these acids.

Report the percent grit of the average of the two determinations to the nearest 0.001 percent.

4.6.3 Width. Use a vernier micrometer clamped in such a manner so that the analyst does not have to hold the micrometer during the test. Condition the sample in accordance with TAPPI T 402. The tape shall be doubled back on itself with the edges parallel and held between the fingers so as to form loop approximately 1 inch in diameter. The loop should not be creased nor its edges handled. One edge of the loop shall be held against the fixed anvil of the micrometer and the loop slowly oscillated back and forth. The micrometer spindle shall be moved up slowly until the first definite increase in resistance is felt in moving the tape. Five determinations, spaced at least 6 inches apart, shall be made. Report the five separate readings and the average width of the roll in inches to the nearest 0.001 inch.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Levels A and B.

Ten rolls of 0.686-inch tape, eight rolls of 0.875-inch, or seven rolls of 1-inch tape of one type, core size, and color shall be prepackaged on a spirally wound, rigid fiber or paperboard tube having an outside diameter not more than 1/8 inch less than the inside diameter of the core. The tube shall be 6-3/4 to 6-7/8 inches long.

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Rolls of tape, packaged as specified above, shall be further packaged flat with the tube vertical in a close fitting box conforming to PPP-B-636, class domestic, style optional. The interior of each box shall be coated with polyethylene or glassine or microcrystalline wax or other suitable greaseproof coating material. Each box shall be closed and sealed by taping all seams, corners, and joints with minimum 3-inch width tape in accordance with method V of the appendix to PPP-B-636.

Alternatively, the rolls of tape, packaged as specified herein, shall be placed in a flexible polyethylene bag constructed from polyethylene film a minimum of 0.0015-inch thick. The seams shall be heat sealed. Closure of the bag shall be accomplished by double-folding the plastic material at the mouth and taping it to the bag body with minimum 1-inch tape, or by eliminating excess air in the bag and heat sealing. Each bag, with the rolls of the flat and the tube vertical, shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic, style optional. The box shall be closed in accordance with method II of the appendix to PPP-B-636.

5.1.2 Commercial. The tape shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the tape against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The rolls of tape, of like description, packaging as specified in 5.1, shall be packed flat with the tubes vertical in a close-fitting box conforming to PPP-B-636, grade V2s, style optional. The weight of the contents of the shipping container shall not exceed the limits of the PPP-B-636. The box shall be closed in accordance with method III, waterproofed in accordance with method V, and banded in accordance with the appendix to PPP-B-636.

5.2.2 Level B. The rolls of tape. of like description, packaged as specified in 5.1, shall be packed flat with the tubes vertical in a close-fitting box conforming to PPP-B-636, class domestic, grade 275, style optional. The weight of the contents of the shipping container shall not exceed the limits of PPP-B-636. The box shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s box fabricated in accordance with PPP-B-636 and closed in accordance with method III of the appendix to PPP-B-636.

5.2.3 Commercial. Teletypewriter perforator tape of one type shall be packed in a fiberboard box in accordance with normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling, and storage, insure delivery at destination, provide for redistribution by the initial receiving activity, and be acceptable by common carrier under the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be as specified in the contract or purchase order (see 6.2).

5.3.2 Military requirements. In addition to any special marking required by the contract or order, intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Teletypewriter perforator tape is used with automatic teletypewriter machines where various combinations of punches operate simultaneously perforating holes in the tape to correspond to teletype signals. Type II tapes are intended for use with optional readers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and, as applicable, color required (see 1.2 and 3.4).
(c) Outside roll diameter and hole diameter in core (see 3.5.1 and 3.5.5).
(d) Minimum length, when required (see 3.5.2).
(e) Width required (see 3.5.3).
(f) Packaging and packing required (see 5.1 and 5.2).
(g) Marking required (see 5.3).
(h) When weather-resistant containers are required for level B packing (see 5.2.2.1).

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